

This modeling approach shows the strengths of fine-tuning a pretrained language model, but also highlights important limitations.

The biggest advantage is that the model does not have to be trained from scratch. DistilBERT has already learned general language patterns from large text corpora, so fine-tuning allows it to adapt quickly to the admissions dataset with relatively little data. This leads to noticeably better performance compared to the earlier neural network (about 77% accuracy versus about 67%), while also removing the need for extensive manual feature engineering.

Another important strength is the combination of text and structured features. Instead of treating GPA, GRE scores, and program names separately, everything is converted into a single text representation. This allows the model to learn relationships between qualitative and quantitative information, such as how certain programs, applicant comments, and scores interact. This is something traditional tabular models struggle to capture.

However, the dataset introduces several sources of bias. GradCafe-style admissions data is self-reported, which means entries may be incomplete, inaccurate, or selectively reported. Strong applicants may be more likely to post results, and certain programs or applicant groups may be overrepresented. Because of this, the model is learning patterns from biased data rather than true admissions behavior.

This also explains why the model can produce misleading or counterintuitive predictions. For example, a strong applicant profile might be predicted as “Rejected” simply because similar profiles in the dataset were rejected or underreported. The model is learning correlations in noisy data, not real-world decision-making processes.

Additionally, the model lacks critical real-world information. Admissions decisions depend on less quantifiable factors such as letters of recommendation, research experience quality, statements of purpose, and institutional priorities. Since we cannot really quantify these, it is hard for the model to be trained to fully simulate what it would be like to apply.

For these reasons, a model like this should not be used for real decision-making. It could easily lead to unfair or misleading conclusions, especially if users interpret predictions as definitive outcomes.

On a public-facing prediction page, uncertainty and user trust must be carefully handled. Even though the model outputs probabilities, users may interpret them as certainty. This is why a clear disclaimer is necessary, and why predictions should be presented as approximate or illustrative rather than definitive. This should especially not be used for people who base their future on things like these, as overconfidence or insecurity may emerge.

Despite these limitations, the “Will You Get In?” project is still very educationally valuable. It demonstrates how modern NLP models are trained and deployed, how structured and unstructured data can be combined, and how real-world datasets introduce noise and bias. It also highlights the importance of ethical considerations when building machine learning systems.